Perfect Roman Domination: Aspects of Enumeration and Parameterization

Kevin Mann(✉) and Henning Fernau

Universität Trier, Fachbereich 4 – Abteilung Informatikwissenschaften, 54286 Trier, Germany
{mann,fernau}@uni-trier.de

Abstract. Perfect Roman dominating functions and unique response Roman dominating functions are two ways to translate *perfect code* into the framework of Roman dominating functions. We also consider the enumeration of minimal perfect Roman dominating functions and show a tight relation to minimal Roman dominating functions. Furthermore, we consider the complexity of the underlying decision problems Perfect Roman Domination and Unique Response Roman Domination on special graph classes. Split graphs are the first graph class for which Unique Response Roman Domination is polynomial-time solvable while Perfect Roman Domination is NP-complete. Beyond this, we give polynomial-time algorithms for Perfect Roman Domination on interval graphs and for both decision problems on cobipartite graphs. However both problems are NP-complete on chordal bipartite graphs. We show that both problems are W[1]-complete if parameterized by solution size and FPT if parameterized by the dual parameter or by clique-width.

1 Introduction

There are many ways to get along with NP-complete problems. For example, you could consider exponential-time algorithms or approximation algorithms. Here, we want to deal with Perfect Roman Domination and Unique Response Roman Domination by looking into enumeration, into the complexity of related decision problems in special graph classes and into parameterized complexity.

We only consider simple undirected graphs $G = (V, E)$ throughout this paper. We recall some notions next. $N_G(v) := \{u \in V \mid uv \in E\}$ describes the *open neighborhood* of $v \in V$ with respect to G. The *closed neighborhood* of $v \in V$ with respect to G is defined by $N_G[v] := N_G(v) \cup \{v\}$. $D \subseteq V$ is a *dominating set* (or a *perfect code*, respectively) of G if each vertex v finds a (unique) vertex u in $D \cap N_G[v]$. A function $f : V \to \{0, 1, 2\}$ is called a *Roman dominating function* (Rdf for short) if each $v \in V$ with $f(v) = 0$ is adjacent to a $u \in V$ with $f(u) = 2$. The weight of a function $f : V \to \{0, 1, 2\}$ is defined by $\omega(f) = \sum_{v \in V} f(v)$. The decision problem Roman Domination has as input a graph G and a positive number k, and the question is if there exists a Rdf such that the weight of the function is at most k. Roman Domination is a well-studied variation of Dominating Set. It is motivated by a defense strategy of the Roman Empire.

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The idea was to position the armies on regions in such a way that either (1) one army is in this region or (2) two armies are on one neighbored region. We model the distribution of the armies on the region (viewed as vertices of the graph) by a function that maps each region to the number of armies placed on this region.

In recent years, notable attention [15,19,20,34,35,38,40,41] was paid to this problem. As for dominating set, there are also many variants of Roman dominating functions considered in the literature. Examples are Roman-{2}-domination (also known as Italian domination) [14] or double Roman domination [1,6,8]. Clearly, as testified by books written on this topic [28,29,31], there are even more variations of the basic concept of domination in graphs.

Here, we will consider two further variations of Roman domination, perfect and unique response Roman domination. A *perfect Roman dominating function* (pRdf for short) (introduced by Henning *et al.* [30]) is a Roman dominating function where each vertex with value 0 has *exactly* one neighbor with value 2. Moreover, if all vertices with at least 1 as a value have no neighbor with value 2, then we have a *unique response Roman dominating function* (urRdf for short) (as introduced in [39]). Both variations can be seen as a way to translate the idea of perfect code into the realm of Rdf. A further motivation can also be the idea of positioning armies on regions: If the armies are placed according to a perfect Rdf and a region without any army is attacked, then it is clear from which region an army moves to secure the attacked region, so no time is wasted to first agree on who is the one to take action and move to the endangered region.

The decision problems are given as follows:

Problem name: Unique Response Roman Domination (URRD)
Given: A graph $G = (V, E)$ and $k \in \mathbb{N}$
Question: Is there a urRdf f on G with $\omega(f) \leq k$?

Problem name: Perfect Roman Domination (PRD)
Given: A graph $G = (V, E)$ and $k \in \mathbb{N}$
Question: Is there a pRdf f on G with $\omega(f) \leq k$?

Let $u_R(G)$ denote the smallest weight of any urRdf f on G. Furthermore, $\gamma_R^p(G)$ denotes the smallest weight of any pRdf on G. The following is known about the complexity of these decision problems. URRD is NP-complete on regular bipartite graphs [9]. Banerjee *et al.* [5] showed that it is NP-complete on chordal graphs, and polynomial-time solvable on distance-hereditary and interval graphs. They also prove that there is no URRD polynomial-time approximation algorithm within a factor of $n^{1-\epsilon}$ for any constant $\epsilon > 0$ and any n-vertex input graph, unless $\mathsf{NP} = \mathsf{P}$. PRD is NP-complete on chordal graphs, planar graphs, and bipartite graphs and polynomial-time solvable on block graphs, cographs, series-parallel graphs, and proper interval graphs, as shown in [7].

Enumeration is a wide area of research, as testified by specialized workshops like [22]. For some examples, we refer to the survey [44]. From a practical point of view, enumeration can be interesting if not all aspects of the problem have been satisfactorily modeled. This makes also sense for the motivation of Roman Domination (variants). It could be the case that between neighbored regions there is a conflict unknown to the strategic planner of the army deployment. Such a conflict could be a reason why the regions do not want to defend each other. Enumerating minimal Rdfs was considered by Abu-Khzam *et al.* [3]. There, it is shown that all pointwise (*i.e.*, for $f, g : V \to \{0, 1, 2\}$, $f \leq g$ iff $f(v) \leq g(v)$ for all $v \in V$) minimal Rdfs of a graph of order n can be enumerated in time $\mathcal{O}(1.9332^n)$ with *polynomial delay.* An enumeration algorithm has *polynomial delay* if the time between two subsequent outputs of the algorithm can be bounded by some polynomial, as well as the time from the start to the first output and the time between the last output and the termination of the algorithm. This property is interesting if a user (or another program) processes the results in a pipeline fashion. Also, it is not known if there is such an algorithm for enumerating (inclusion-wise) minimal dominating sets. In this context, we remark:

Observation 1. *Unless* $\mathsf{P} = \mathsf{NP}$*, there exists no polynomial-delay enumeration algorithm for (inclusion-wise minimal) perfect codes.*

Namely, such a polynomial-delay enumeration algorithm could also be used to decide (in polynomial time) if a perfect code exists, which is NP-hard, see [33].

We now collect more notation that we will use in the paper. Let $\mathbb{N}$ denote the set of all nonnegative integers (including 0). For $n \in \mathbb{N}$, we will use the notation $[n] := \{1, \ldots, n\}$. Let A, B be two sets. B^A is the set of all function $f : A \to B$. For $A \subseteq B$, $\chi_A : B \to \mathbb{N}$ denotes the characteristic function ($\chi_A(a) = 1$ iff $a \in A$; $\chi_A(a) = 0$, otherwise). Let $G = (V, E)$ be a graph. For a set $A \subseteq V$, the open neighborhood is defined as $N_G(A) := \left(\bigcup_{v \in A} N_G(v)\right)$. The closed neighborhood of A is given by $N_G[A] := N_G(A) \cup A$. The *private neighborhood* of a vertex $v \in A$ with respect to G and A is denoted by $P_{G,A}(v) := N_G[v] \setminus N_G[A \setminus \{v\}]$. Let $G = (V, E)$ be a graph and $f \in \{0, 1, 2\}^V$. We define $V_i(f) := \{v \in V \mid f(v) = i\}$ for $i \in \{0, 1, 2\}$. If $v \in V$ obeys $f(v) = 2$, then $u \in N_G(v)$ with $f(u) = 0$ is called a *private neighbor* of v (with respect to f) if $|N_G(u) \cap V_2(f)| = 1$.

Observation 2. *Let* $G = (V, E)$ *be a graph and* $f : V \to \{0, 1, 2\}$*. (1) If* f *is a pRdf, then for all* $v \in V_2(f)$ *and* $u \in N(v) \cap V_0(f)$*,* u *is a private neighbor of* v*. (2) If* f *is a Rdf such that for all* $v \in V_2(f)$ *and* $u \in N(v) \cap V_0(f)$*,* u *is a private neighbor of* v*, then* f *is a pRdf.*

Observation 3. *Each urRdf on a graph without isolates is a minimal Rdf.*

A final note on parameterized complexity: If P is some problem and κ is some parameter, κ-P denotes the problem P parameterized by κ, see [18, 24].

2 Enumerating Minimal Perfect Roman Domination

The main goal of this section is to find a polynomial-delay algorithm to enumerate all minimal pRdfs even when solving the extension problem is hard, which blocks the standard approach to getting such enumeration algorithms by a branching algorithm that checks extendibility and prunes unnecessary branches.

Problem name: EXTENSION PERFECT ROMAN DOMINATION (EXT PRD)
Given: A graph $G = (V, E)$ and $f \in \{0, 1, 2\}^V$.
Question: Is there a (pointwise) minimal pRdf g on G with $f \leq g$?

Theorem 1. *[23]* Ext PRD *is* NP*-complete.* $\omega(f)$-Ext PRD *is* W[1]*-complete and* $|V_0(f)|$-Ext PRD *is* W[2]*-complete (even on bipartite graphs).*

We often use the following characterization result that is of independent interest.

Theorem 2. *Let* $G = (V, E)$ *be a graph. A function* $f : V \to \{0, 1, 2\}$ *is a minimal pRdf iff the following conditions are true:*

1. $\forall v \in V_0(f) : |N_G(v) \cap V_2(f)| = 1,$
2. $\forall v \in V_1(f) : |N_G(v) \cap V_2(f)| \neq 1,$
3. $\forall v \in V_2(f) : |N_G(v) \cap V_0(f)| \neq 0.$

Proof. Let f be a minimal pRdf. This implies the first condition. If there exists a $v \in V_1(f)$ with $|N_G(v) \cap V_2(f)| = 1$, then $f - \chi_{\{v\}}$ is also a pRdf, cf. Observation 2. For each $v \in V_2(f)$, $|N_G(v) \cap V_0(f)| \neq 0$, as otherwise $f - \chi_{\{v\}}$ is also a pRdf.

Now assume f is a function that fulfills these conditions. By Condition 1, we know f is a pRdf. Let g be a minimal pRdf with $g \leq f$. Thus, $V_0(f) \subseteq V_0(g)$ and $V_2(g) \subseteq V_2(f)$ hold. Assume there exists a $v \in V$ with $g(v) < f(v) = 2$. By Condition 3, there exists some $u \in N_G(v) \cap V_0(f)$. Because of condition 1, v is the only neighbor of u with value 2. Hence, $N_G(u) \cap V_2(g) \subseteq N_G(u) \cap (V_2(f) \setminus \{v\}) = \emptyset$. This contradicts the construction of g as a pRdf. Therefore, $V_2(g) = V_2(f)$. If there were a $v \in V$ with $g(v) = 0 < 1 = f(v)$, then this contradicts that g is a pRdf, since $|N_G(v) \cap V_2(g)| = |N_G(v) \cap V_2(f)| \neq 1$. □

Remark 1. This result implies that the constant function χ_V is the maximum minimal pRdf with respect to ω for each graph $G = (V, E)$, as for Rdfs, see [13].

We use Theorem 2 also to obtain the announced polynomial-delay enumeration algorithm. This algorithm uses the idea that there is a bijection between all minimal pRdfs and all minimal Rdfs of a graph. To show this bijection, we need the following characterization theorem from [2].

Theorem 3. *[2] Let* $G = (V, E)$ *be a graph,* $f : V \to \{0, 1, 2\}$ *be a function and let* $G' := G[V_0(f) \cup V_2(f)]$*. Then,* f *is a minimal Rdf iff the following holds:*
(1) $N_G[V_2(f)] \cap V_1(f) = \emptyset$*, (2)* $\forall v \in V_2(f) : P_{G',V_2(f)}(v) \nsubseteq \{v\}$*, and (3)* $V_2(f)$ *is a minimal dominating set of* G'*.*

Let $G = (V, E)$ be graph. Define the sets of functions
$\mu - \mathcal{RDF}(G) := \{ f : V \to \{0, 1, 2\} \,|\, f \text{ is a minimal RDF} \}$ and
$\mu - \mathcal{PRDF}(G) := \{ f : V \to \{0, 1, 2\} \,|\, f \text{ is a minimal pRdf} \}$.

Theorem 4. *Let* $G = (V, E)$ *be graph. There is a bijection* $B : \mu - \mathcal{RDF}(G) \to \mu - \mathcal{PRDF}(G)$*. Furthermore,* $B(f)$ *and* $B^{-1}(g)$ *can be computed in polynomial time (with respect to* G*) for each* $f \in \mu - \mathcal{RDF}(G)$ *and* $g \in \mu - \mathcal{PRDF}(G)$*.*

Proof. Let $f \in \mu - \mathcal{RDF}(G)$ and $g \in \mu - \mathcal{PRDF}(G)$. Define $B(f)$ by the sets

$$V_0(B(f)) = \{v \in V_0(f) \mid |N_G(v) \cap V_2(f)| = 1\},$$
$$V_1(B(f)) = \{v \in V_0(f) \mid |N_G(v) \cap V_2(f)| \geq 2\} \cup V_1(f), \quad V_2(B(f)) = V_2(f).$$

By definition of B, $|N_G(v) \cap V_2(f)| = 1$ holds for all $v \in V_0(B(f))$. Since each $v \in V$ with $f(v) = 2$ needs a private neighbor except itself by Theorem 3, $N_G(v) \cap V_0(B(f)) \neq \emptyset$ holds. As any $v \in V_1(f)$ has no neighbor in $V_2(f)$ and any $v \in V_1(B(f)) \setminus V_1(f)$ has at least two neighbors in $V_2(f)$, all conditions of Theorem 2 are met. Therefore, $B(f) \in \mu - \mathcal{PRDF}(G)$. Define $B^{-1}(g)$ by

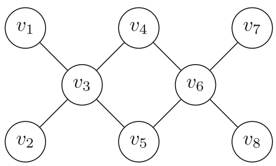
$$V_0(B^{-1}(g)) = V_0(g) \cup \{v \in V_1(g) \mid |N_G(v) \cap V_2(g)| \geq 2\},$$
$$V_1(B^{-1}(g)) = \{v \in V_1(g) \mid |N_G(v) \cap V_2(g)| = 0\}, \quad V_2(B^{-1}(g)) = V_2(f).$$

By definition of B^{-1}, $N_G(v) \cap V_2(B^{-1}(g)) = \emptyset$ for each $v \in V_1(B^{-1}(g))$. By Theorem 2, each $v \in V_0(g)$ has exactly one neighbor in $V_2(g)$. Hence, each vertex in $V_0(B^{-1}(g))$ has a neighbor in $V_2(B^{-1}(g))$ and $V_2(B^{-1}(g))$ is a dominating set of $G[V_0(B^{-1}(g)) \cup V_2(B^{-1}(g))]$. By Theorem 2, each $v \in V_2(g)$ has a $u \in N_G(v) \cap V_0(g)$ with $\{v\} = N_G(u) \cap V_2(g)$. So, u is a private neighbor of v and $B^{-1}(g) \in \mu - \mathcal{RDF}(G)$. We leave to the reader to prove: B is bijective. ☐

This bijection is a so-called *parsimonious reduction*. This is a class of reductions designed for enumeration problems. For more information, we refer to [42]. Even without this paper, Theorem 4 implies some further results thanks to [2].

Corollary 1. *There are graphs of order n that have at least $\sqrt[5]{16}^{\,n} \in \Omega(1.7441^n)$ many minimal pRdfs.*

Theorem 5. *There is a polynomial-space, polynomial-delay algorithm that enumerates all minimal pRdfs of a given graph of order n in time $\mathcal{O}^*(1.9332^n)$.*



$f = 2\chi_{\{v_3,v_6\}}$ is a minimum Rdf ($\omega(f) = 4$) and $g = 2\chi_{\{v_3\}} + \chi_{\{v_6,v_7,v_8\}}$ is a minimum pRdf ($\omega(g) = 5$), but $\omega(B(f)) = 6$, $\omega\left(B^{-1}(g)\right) = 5$.

Fig. 1. Counter-example of Remark 2

Remark 2. The function B defined in the proof of Theorem 4 inherits inclusion-wise minimality, but this does not mean that, if f is a minimum Roman dominating function, then $B(f)$ is a minimum pRdf. Consider Fig. 1.

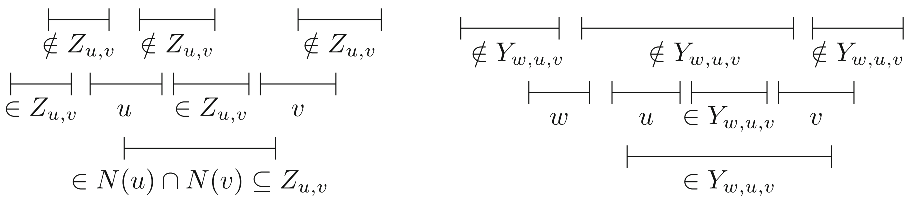
According to [4], Theorem 4 implies some further results for special graph classes. Let $G = (V, E)$, with $n := |V|$ being the order of G. If G is a forest or interval graph, then there is a pRdf enumeration algorithm that runs in time $\mathcal{O}^*(\sqrt{3}^{\,n})$ with polynomial delay. For both graph classes, a graph with many isolated edges is example of a graph with $\sqrt{3}^{\,n}$ many minimal pRdfs. This is also the worst-case example for chordal graphs which is known so far. We can enumerate all pRdfs of these graphs in time $\mathcal{O}(1.8940^n)$. If G is a split or cobipartite graph, then we can enumerate all pRdfs of G in $\mathcal{O}^*(\sqrt[3]{3}^n)$ with polynomial delay. Further, both classes include graphs of order n with $\Omega(\sqrt[3]{3}^n)$ many pRdfs. There is also a polynomial-time recursive algorithm to count all pRdfs of paths. Enumeration of urRdfs in split graphs is considered in Subsect. 3.4.

3 Complexity and Enumeration in Special Graph Classes

In this section, we will consider the complexity of our problems PRD and URRD on different graph classes: interval, split, cobipartite and chordal bipartite.

3.1 Interval Graphs

We first start with a polynomial-time algorithm for PRD for interval graphs. A graph $G = (V, E)$ is called an *interval graph* (see [25]) if, for $v \in V$, there is an interval $I_v = [l_v, r_v]$ such that $\{v, u\} \in E$ iff $I_v \cap I_u \neq \emptyset$. We can assume that the interval representation obeys $\forall u, v \in V : r_u = r_v \implies u = v$.



(a) Illustrating $Z_{u,v}$. (b) Illustrating $Y_{w,u,v}$.

Fig. 2. Examples illustrating our notions.

Theorem 6. PRD *is polynomial-time solvable on interval graphs.*

Proof. Let $G = (V, E)$ be an interval graph such that $I_v = [l_v, r_v]$ is the interval representation of $v \in V$. The dynamic programming works from "left to right" according to some linear extension of the partial order defined as $w \leq v$ iff $l_w \leq l_v \wedge r_w \leq r_v$. For $v \in V$, $L_v \coloneqq \{w \in V \mid l_w \leq r_v\}$ denotes the set of vertices which are dominated by v or "to the left" of I_v. To simplify our notation, we define $P_v \coloneqq \{w \in V \setminus \{v\} \mid w \leq v\}$ to easily refer to the already considered vertices in the dynamic program. $W_v \coloneqq \{w \in V \mid r_v < r_w\}$ is the set of not yet considered vertices. For $v \in V$, $u \in P_v$ and $w \in P_u$, define $Z_{u,v} \coloneqq (L_v \setminus (N[u] \cup N[v])) \cup (N(u) \cap N(v))$ and $Y_{w,u,v} \coloneqq ((N(u) \cap N(v)) \setminus N[w]) \cup (L_v \setminus (L_u \cup N[v]))$, see Fig. 2. All these sets and their cardinalities can be computed before the actual dynamic programming starts. Also, we can compute the pRdfs that set at most one vertex to 2 and store their minimum weight in $\min_{\leq 1}$.

We need some more notation to formulate the dynamic programming algorithm and prove its correctness. For $v \in V$ and $u \in P_v$, define $\mathit{\Gamma}_{u,v} := \{f \in \{0,1,2\}^V \mid f|_{L_v} \in \mathcal{PRDF}\,(G[L_v]) \wedge f(u) = f(v) = 2 \wedge \forall w \in W_u \setminus \{v\} : f(w) \neq 2\}$ and $\gamma_{u,v} = \min_{f \in \mathit{\Gamma}_{u,v}} \omega\,(f)$. In the following, $f'_{u,v}$ will denote one function from $\mathit{\Gamma}_{u,v}$ that attains this minimum, i.e., $\omega\left(f'_{u,v}\right) = \gamma_{u,v}$. For the inductive computation, we will reuse the values $\gamma_{w,u}$ with $w \in P_u$. For the correctness of this approach, the following observations are helpful.

1. $Z_{u,v}$ is tied to the function $f_{u,v} \in \mathit{\Gamma}_{u,v}$ by $V_2(f_{u,v}) = \{u, v\}$, $V_1(f_{u,v}) = Z_{u,v}$.
2. $Y_{w,u,v} = V_1(f_{w,u,v}) \setminus V_1(f'_{w,u})$, where $f_{w,u,v}$ is the auxiliary pRdf which we will construct from $f'_{w,u}$ in the induction step.
3. If $f \in \mathit{\Gamma}_{u,v}$, then $\tilde{f} := f + \chi_{V \setminus L_v}$ is a pRdf on G. If f is minimum, then $\tilde{f}$ is a minimum pRdf on G.

By the third property, after the inductive computations, $\min\{\min_{\leq 1}, \min\{\gamma_{u,v} + |V \setminus L_v| \mid v \in V, u \in P_v\}\}$ gives the minimum weight of any pRdf for G.

Assume now that v is a vertex for which we want to compute all relevant $\gamma_{u,v}$ next, $u \in P_v$. The following claim contains a formula that describes how to do this. It implicitly also covers the initial step: then $\gamma_{u,v} = 4 + |Z_{u,v}|$ since $P_u = \emptyset$.

Claim. Let $v \in V$ and $u \in P_v$. Then

$$\gamma_{u,v} = \min\{4 + |Z_{u,v}|, 2 - |\{v\} \cap N(u) \cap N(w)| + \gamma_{w,u} + |Y_{w,u,v}| \mid w \in P_u\}.$$

Formal correctness arguments are omitted due to space constraints. □

3.2 Split Graphs

A *split graph* $G = (V, E)$ is a graph in which the vertex set V can be partitioned into C, I, where C is a clique and I is an independent set of G. It seems to be open to which complexity class PRD and URRD belong on split graphs, as explicitly asked in [5]. We will answer this question in a somewhat surprising manner: while URRD can be solved in linear time, PRD turns out to be NP-complete. First, we will consider urRdfs on split graphs and prove a combinatorial result.

Lemma 1. *Let $G = (V, E)$ be a connected split graph with a clique C and an independent set I such that $V = C \cup I$ and $C \cap I = \emptyset$. For each urRdf $f : V \to \{0,1,2\}$, one of the following conditions holds: (1) $V_2(f) \cap I = \emptyset$ and $|V_2(f) \cap C| \leq 1$; or (2) $V_2(f) \subseteq I$.*

Proof. Suppose that $V_2(f) \nsubseteq I$. By the definition of urRdf, two vertices of C cannot have the value 2, since C is a clique. Hence, $|V_2(f) \cap C| \leq 1$. For the sake of contradiction, assume there are $v \in I$ and $u \in C$ with $f(u) = f(v) = 2$. As f is a urRdf, u and v cannot be neighbors. Since G is connected, $\emptyset \subsetneq N_G(v) \subseteq C \setminus \{u\} \subseteq N_G(u)$. This contradicts the assumption that f is a urRdf. □

This lemma implies that we can solve URRD by looking for a vertex v of maximum degree (which must be in an inclusion-wise maximal clique), set the value of v to two and the values of the other vertices accordingly. This implies:

Theorem 7. (∗) URRD *can be solved in linear time on split graphs.*

Interestingly, the seemingly very similar problem PRD on split graphs is harder to solve. To see this, we reduce from a problem called PERFECT CODE.

Theorem 8. (∗) PRD *on split graphs is* NP*-complete.*

We contrast this by giving a branching algorithm for PRD. Our algorithm relies on combinatorial insights from Theorem 4 and [4], where Rdfs have been discussed on split graphs, so that $V_2(f')\cap I = \emptyset$ or $V_2(f')\cap C = \emptyset$ for minimum f'.

Theorem 9. (∗) $\omega(f) -$ PRD *can be solved in time* $\mathcal{O}^*(\varphi^{\omega(f)})$ *on split graphs, where* φ *is the golden ratio.*

3.3 Cobipartite Graphs

Now, we consider *cobipartite graphs*, the complementary graphs of bipartite graphs, i.e., the vertex set can be partitioned into two cliques. On this graph class, PRD as well as URRD are solvable in polynomial time.

Theorem 10. PRD *is polynomial-time solvable on cobipartite graphs.*

Proof. Let $G = (V, E)$ be a cobipartite graph, with V partitioned into the two cliques $C_1, C_2 \subseteq V$. For $v \in V$, define the pRdf$g_v := 2 \cdot \chi_{\{v\}} + \chi_{V\setminus N[v]}$.

Let $f \in \{0,1,2\}^V$ be a pRdf with $|V_2(f) \cap C_1| \geq 2$ and $v \in C_2$. Next, we prove that $\omega(g_v) \leq \omega(f)$. For each $u \in C_1 \setminus V_2(f)$, $|V_2(f) \cap N(u)| \geq 2$. Hence $C_1 \subseteq V_1(f) \cup V_2(f)$. Since $C_2 \subseteq N[v] = V_0(g_v) \cup \{v\}$, $\omega(g_v) \leq |C_1| + 2 \leq \omega(f)$. Symmetrically, $\omega(g_v) \leq |C_1| + 2 \leq \omega(f)$ for $v \in C_1$ and a pRdf f with $|V_2(f) \cap C_2| \geq 2$. Let f be a minimum pRdf on G. By the previous thoughts, we can assume $|V_2(f) \cap C_i| \leq 1$ for each $i \in \{1, 2\}$. This leaves only $|V|^2 + |V|$ many possibilities to check. □

Cabrera *et al.* [9] proved that URRD is NP-complete on bipartite graphs. For cobipartite graphs, this problem is polynomial-time solvable. For this result, we use the idea of 2-packings. Let $G = (V, E)$ be a graph. A set $S \subseteq V$ is called a *2-packing* if the distance between any two vertices in S is at least 3. Targhi *et al.* [43] presented a proof for $u_r(G) = \min\{2|S| + |V(G) \setminus N_G[S]| \mid S \text{ is a 2-packing}\}$. Analogously to this, we can prove that, for each graph $G = (V, E)$, there exists a bijection ψ_G between all 2-packings of the graph and all urRdfs. Here, for a 2-packing S and for $x \in V$, $\psi_G(S) := 2 \cdot \chi_S + \chi_{V\setminus N(S)}$. This also yields that, for each urRdf f, $V_2(f)$ is a 2-packing. This is the main idea of our algorithm.

Lemma 2. (∗) URRD *is polynomial-time solvable on cobipartite graphs. Furthermore, there are at most* $\frac{|V|^2}{4}$ *many 2-packings, or urRdfs, in* $G = (V, E)$.

The upper bound $\frac{|V|^2}{4}$ is tight: consider the complement of the complete bipartite graph $K_{t,t}$, where both classes have the same size t. This is interesting as there could be exponentially many urRdfs even on connected split graphs, which is quite a related class of graphs that we consider next.

3.4 Remarks on Enumeration of urRdfs in Split Graphs

Enumerating urRdfs in general graphs in time $\mathcal{O}(1.5399^n)$ was shown in [32].

Corollary 2. *All urRdfs of a connected split graph of order n can be enumerated in time $\mathcal{O}(\sqrt[3]{3}^n) \subseteq \mathcal{O}(1.4423^n)$.*

Proof. Non-trivial connected split graphs have no isolates. By Observation 3, we can hence use the enumeration algorithm for minimal Rdfs on split graphs of order n from [4] which runs in time $\mathcal{O}(\sqrt[3]{3}^n)$. □

$\mathcal{O}(\sqrt[3]{3}^{|V|})$ is even a tight upper bound for connected split graphs as shown next.

Theorem 11. *There is an infinite family of connected split graphs $G_t = (V_t, E_t)$ with $\Omega(\sqrt[3]{3}^{|V_t|})$ many urRdfs.*

Proof. We only need to consider the split graph family $G_t = (V_t, E_t)$ with $V_t := C_t \cup I_t$, $C_t := \{c_1, \ldots, c_t\}$, $I_t := \{v_1, \ldots, v_{2t}\}$, $E_t := \binom{C_t}{2} \cup \{\{c_i, v_{2i-1}\}, \{c_i, v_{2i}\} \mid i \in \{1, \ldots, t\}\}$. Clearly, $|V_t| = 3t$. By arguments from Theorem 7, for each urRdf f on G_t with $|V_2(f) \cap C_t| = 1$, in fact $|V_2(f)| = 1$. There are t many such urRdfs. Let f be a urRdf with $V_2(f) \cap C_t = \emptyset$. In this situation, for each $i \in \{1, \ldots, t\}$, there are three ways to dominate c_i: $f(c_i) = f(v_{2i-1}) = f(v_{2i}) = 1$, $f(c_i) = f(v_{2i}) = 0$ and $f(v_{2i-1}) = 2$, or $f(c_i) = f(v_{2i-1}) = 0$ and $f(v_{2i}) = 2$. Hence, there are $t + 3^t = \frac{|V|}{3} + \sqrt[3]{3}^{|V|}$ many urRdfs. □

For general graphs, the best lower bound [32] is $\mathcal{O}(1.4970^n)$.

The polynomial-delay property of the Rdf-enumeration algorithm from [4] is not inherited for enumerating urRdf, as not each minimal Rdf is a urRdf. Nonetheless, we will present a sketch of a direct polynomial-delay branching enumeration algorithm for enumerating urRdfs that is optimal due to Theorem 11. For this purpose, we consider each 2-packing on a connected split graph and use the bijection between urRdfs and 2-packings. Our algorithm is a modification of the recursive backtracking algorithm of [37] and has polynomial delay with the same arguments. The algorithm of [37] is a general approach to enumerate all sets of $\mathcal{F} \subseteq 2^U$ for a universe U, where $\mathcal{F}$ fulfills the following downward closure property: if $X \in \mathcal{F}$ and $Y \subseteq X$, then $Y \in \mathcal{F}$.

Theorem 12. $(*)$ *All urRdfs of a connected split graph of order n can be enumerated in time $\mathcal{O}(\sqrt[3]{3}^n)$ with polynomial delay and polynomial space.*

One of the rather surprising aspects of the omitted proof is that we do not need any sophisticated measure-and-conquer methodology to analyze the runtime of the branching algorithm by a simple case analysis concerning the vertex degrees. In the core part of the algorithm, we decide if a vertex in the independent set is put into the related 2-packing or not, also cf. [37].

3.5 Chordal Bipartite Graphs

These are bipartite graphs where each induced cycle has length 4. To prove NP-hardness of PRD and URRD on chordal bipartite graphs, we use the problem One-in-Three 3SAT, with a reduction inspired by [36, Theorem 2.2] for the NP-hardness of PC. However, our proof is much more sophisticated in order to deal with the weights of the pRdfs.

Theorem 13. (∗) PRD& URRD *are NP-complete on chordal bipartite graphs.*

4 Parameterized Complexity

In this section we, consider parameterized complexity aspects of PRD and URRD. More precisely, we consider the parameters *clique width* (cwd for short, defined in [16]), solution size and its dual, i.e., order minus solution size.

First, we give a LinEMSOL(τ_1) formula for the optimization problems corresponding to PRD and to URRD in order to show that these are linear-time FPT if parameterized by cwd, using now classical results from [17,26]. cwd is an interesting parameter, since an FPT algorithm with cwd as parameter implies FPT algorithms with respect to tw, nd or vc as parameters. However, the algorithmic results of the previous section do not follow from these considerations, as it is known that even unit interval graphs have unbounded cliquewidth, see [26].

Lemma 3. Minimum PRD *on the graph $G = (V, E)$ can be expressed as*

$$\underset{X_1 \subseteq V, X_2 \subseteq V}{\operatorname{argmin}} \{|X_1| + 2|X_2| \mid \langle G, X_1, X_2 \rangle \vDash \theta(X_1, X_2)\}, \text{ with } \theta(X_1, X_2) \text{ equal to}$$

$$\forall v \,(X_1(v) \vee X_2(v) \vee \exists u \,[X_2(u) \wedge E(v, u) \wedge \forall w \,([X_2(w) \wedge E(w, v)] \Rightarrow u = w)]) \,.$$

Proof. Let $G = (V, E)$ be a graph for which we want to determine the pRdf of smallest weight. First, if f is some pRdf of G, then $\theta(V_1(f), V_2(f))$ is obviously true. Now we want to show why $X_1, X_2 \subseteq V$ as a minimum solution would describe a pRdf. Define $f := \chi_{X_1} + 2\chi_{X_2}$. We can then see that for all $v \in V$, $v \in X_1$, $v \in X_2$, or there exists a $u \in X_2 \cap N(v)$ such that there is no other neighbor of v in X_2. Therefore, we only need to show $X_1 \cap X_2 = \emptyset$. So, assume there is a $v \in X_1 \cap X_2$. Then for $X_1' := X_1 \setminus \{v\}$, (X_1', X_2) is a smaller solution. Hence, f is a pRdf. □

Theorem 14. *cwd* − PRD ∈ *FPT.*

By [26], distance-hereditary graphs have cliquewidth bounded by three, so that we get the following supplement to the previous section.

Corollary 3. PRD *can be solved in linear time on distance-hereditary graphs.*

With similar techniques, we can also prove an analogous result for URRD.

Theorem 15. *cwd* − URRD ∈ *FPT.*

Beside structural parameters, also the solution size is an interesting often-used parameter. We show next that $\omega(f)$-URRD and $\omega(f)$-PRD are W[1]-complete. We do this by using Steps-Short Non-Deterministic Turing Machine Computation (SNTMC) which is known to be W[1]-complete (see [10–12,18]), as well as PC, parameterized by solution size. So, we cannot expect FPT-results. We start our presentation with a W[1]-membership proof, which is in a mathematical sense the most interesting proof. As a side-product, we also get membership in XP. To this end, we need the following interesting structural characterization lemma that is proven by employing some double-counting arguments.

Lemma 4. (∗) *Let $G = (V, E)$ be a graph and $f \in \{0, 1, 2\}^V$ be a function such that $N(u) \cap N(v) \cap V_0(f) = \emptyset$ for all $u, v \in V_2(f)$, $u \neq v$. Then f is a pRdf iff*

$$|V| = \left(\sum_{u \in V_1(f)} (1 - |N(u) \cap V_2(f)|)\right) + \left(\sum_{v \in V_2(f)} (|N[v]| - |N(v) \cap V_2(f)|)\right).$$

Theorem 16. $\omega(f)$-PRD *belongs to* W[1].

Proof. (Sketch) We will use Steps-SNTMC. So let $G = (V, E)$ be a graph of order n and $k \in \mathbb{N}$. The reduction will compute a Turing machine M with tape alphabet containing $V \cup \{\zeta\}$, with $\zeta \notin V$. In its internal memory, M stores a table T that contains, for $u, v \in V$, $u \neq v$, either $N(u) \cap N(v)$ (if $|N(u) \cap N(v)| \leq k$) or ∞. Moreover, the reduction machine also computes addition and subtraction tables for numbers up to n and the alphabet contains also these numbers, so that M can do basic arithmetics, like reading two number symbols from the tape and printing the result, in constant time. Finally, M has access to a table of the degrees of all vertices in its internal memory. The Turing machine M will work in 5 phases as follows:

1. Guess the vertices in $V_2(f)$ and then in $V_1(f)$ and write them (as symbols) on the tape. We separate the vertices via the symbol ζ. This takes at most $k+1$ many steps.
2. Also, in time $\mathcal{O}(k)$ M can check if twice the number of V_2-symbols (left of the separator ζ) plus the number of V_1-symbols (right to ζ) is at most k. If not, M stops in a non-final state.
3. Check if we enumerated any vertex in $V_1(f) \cup V_2(f)$ twice (in $\mathcal{O}(k^2)$ steps). If this is the case, then the machine stops in a non-final state.
   Otherwise, we know now that the V_1-symbols together with the V_2-symbols that we guess do define a function f of weight $\omega(f) \leq k$.
4. Check for each pair of vertices $u, v \in V_2(f)$ $(u \neq v)$ if $N(u) \cap N(v) \subseteq V_1(f) \cup V_2(f)$. This condition is clearly necessary if f should be a pRdf. For this check, we need table T. If $T[u, v] = \infty$, $N(u) \cap N(v) \subseteq V_1(f) \cup V_2(f)$ is impossible and we can stop in a non-final state. Otherwise, M checks element by element of $N(u) \cap N(v)$ if it is in $V_1(f) \cup V_2(f)$. Altogether, this needs at most $\mathcal{O}(k^3)$ many steps.

5. For each $v \in V_1(f) \cup V_2(f)$, M computes $|N(v) \cap V_2(f)|$ (at most $\mathcal{O}(k^2)$ steps). Use these numbers to compute (with basic arithmetics)

$$n' := \left(\sum_{u \in V_1(f)} (1 - |N(u) \cap V_2(f)|) \right) + \left(\sum_{v \in V_2(f)} (|N[v]| - |N(v) \cap V_2(f)|) \right)$$

Finally, M uses Lemma 4 to check if f is a pRdf, i.e., if $n = n'$. □

Using a construction simpler than the previous one for W[1]-membership together with a relatively straightforward reduction from PC for the W[1]-hardness that basically introduces $2k$ twins per vertex, we can prove:

Theorem 17. (∗) $\omega(f)$-URRD *is* W[1]*-complete.*

The previous W[1]-hardness result could also be used to show that $\omega(f)$-PRD is W[1]-complete. We can show this type of result even on bipartite graphs.

Theorem 18. (∗) $\omega(f)$-PRD *is* W[1]*-complete even on bipartite graphs.*

Different from PC, it makes sense to consider the dual parameterization $(|V| - k)$, as it is even NP-hard to find out if the graph has any perfect code (see [33]). So, the problem is para-NP-hard for the dual parameterization. This is not the case for PRD and URRD, since the constant 1 function is always a pRdf/urRdf. Therefore, these problems are trivially solvable if $n - k = 0$. We could even see this parameter as a *distance from triviality* parameter as in [27].

Theorem 19. $(n - k) -$ PRD, $(n - k) -$ URRD $\in$ FPT.

Proof. Let $G = (V, E)$ be graph and $f \in \{0, 1, 2\}^V$. Then

$$\begin{aligned} n - \omega(f) &= \sum_{v \in V} (1 - f(v)) \\ &= \left(\sum_{v \in V_0(f)} 1 - 0 \right) + \left(\sum_{v \in V_1(f)} 1 - 1 \right) + \left(\sum_{v \in V_2(f)} 1 - 2 \right) \\ &= |V_0(f)| - |V_2(f)|. \end{aligned}$$

We will use this equality to bound the diameter and degree by $n - k$.

Assume there is a vertex v with $\deg(v) \geq n - k + 1$. Then $f_v := \chi_V + \chi_{\{v\}} - \chi_{N(v)}$ is clearly a urRdf and therefore a pRdf with

$$n - \omega(f_v) = |V_0(f_v)| - |V_2(f_v)| = \deg(v) - 1 \geq n - k.$$

Hence, $\omega(f_v) \leq k$ and we observe a yes-instance and can hence use the consideration of a vertex of maximum degree as a reduction rule.

Assume there is a shortest path $p = v_1, \ldots v_{3 \cdot (n-k)}$ of length $3 \cdot (n - k)$. Define $f_p := \chi_V + \sum_{i=0}^{n-k-1} (\chi_{\{v_{3i+2}\}} - \chi_{N(v_{3i+2})})$. Clearly, each vertex in $V_0(f)$

has at least one neighbor in $V_2(f)$. Assume there exists a vertex $u \in V_0(f)$ with $v_{3i+2}, v_{3j+2} \in V_2(f) \cap N(u)$ with $i \neq j$ (w.l.o.g., $i < j$). This would imply that $v_1, \ldots, v_{3i+2}, u, v_{3j+2}, \ldots, v_{3\cdot(n-k)}$ is a shorter path. Hence, f_p is a pRdf. Analogously, $V_2(f_p)$ is an independent set. Therefore, f_p is even a urRdf. Further,

$$n - \omega\left(f_p\right) = |V_0(f_p)| - |V_2(f_p)| \geq 2 \cdot (n-k) - n - k \geq n - k.$$

So, $\omega\left(f\right) \leq k$ and we observe a yes-instance and can hence use the consideration of the graph diameter as a reduction rule. As maximum degree and diameter are bounded in $n-k$, the number of vertices is bounded by the *Moore bound*, giving $\mathcal{O}\left((n-k)^{3\cdot(n-k)}\right)$ as the kernel size, see [21]. □

5 Final Remarks

We have enhanced the understanding of the combinatorial structure and the complexity of PRD and URRD in many ways. But many aspects are still open to explore. For instance, we did not touch the question of kernel sizes in any of our FPT results. Also, the fact that the 'bijection approach' that we took for enumeration necessitates parsimonious reductions could be used towards investigating counting complexities, again an untouched ground in this area.

We employed (∗) when proofs have been omitted due to space constraints.

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